

LAHR, Germany

WMO: 108050

Lat: 48.3647N

Lon: 7.8281E

Elev: 515

StdP: 14.42

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	20.4	11.1	9.9	18.0	15.7	12.4	23.4	44.1	50.6	31.7	46.3	4.4	20	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	87.8	69.5	84.3	68.4	81.1	67.1	71.3	83.4	69.8	81.2	68.4	78.6	5.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.5	103.1	75.8	66.0	97.8	74.3	64.6	93.0	72.7	35.4	83.6	34.1	81.3	32.9	78.6	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
20.3	16.6	14.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			9.3	93.8	5.1	3.2	5.6	96.1	2.6	98.0	-0.3	99.8	-4.0	102.1
			8.6	75.0	5.1	1.9	5.0	76.3	2.0	77.4	-0.8	78.5	-4.5	79.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	35.4	38.1	44.3	51.4	59.2	65.0	67.7	67.4	59.8	52.4	42.8	37.0	(6)
	DBStd	13.50	8.77	8.25	7.29	6.70	6.36	6.23	5.36	5.37	5.70	7.00	7.07	8.24	(7)
	HDD50	1761	456	336	203	63	5	0	0	0	1	57	232	408	(8)
	HDD65	5169	918	752	643	408	200	78	32	34	174	394	667	869	(9)
	CDD50	2412	3	4	26	105	290	451	549	541	295	130	15	3	(10)
	CDD65	345	0	0	0	0	20	79	116	110	18	2	0	0	(11)
	CDH74	3189	0	0	0	35	247	727	1040	960	165	15	0	0	(12)
	CDH80	939	0	0	0	1	48	205	340	318	26	1	0	0	(13)
(14) Wind	WSAvg	5.6	6.6	7.1	6.3	6.0	5.7	5.1	5.0	4.6	4.7	4.9	5.2	6.1	(14)
(15) Precipitation	PrecAvg	35.2	2.1	2.2	2.3	2.6	3.6	3.6	3.9	3.2	2.9	3.1	3.0	2.8	(15)
	PrecMax	44.1	4.8	4.1	6.7	5.1	6.5	7.0	11.4	7.3	5.1	6.6	6.2	5.7	(16)
	PrecMin	25.1	0.4	0.6	0.5	0.2	0.9	1.1	1.1	0.5	1.1	0.3	0.1	0.2	(17)
	PrecStd	4.5	1.2	1.0	1.4	1.3	1.5	1.5	2.0	1.5	1.2	1.4	1.4	1.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	60.3	69.3	78.1	85.5	90.6	92.1	94.6	83.6	75.6	62.9	57.9	(19)
		MCWB	51.3	49.3	53.4	59.5	66.0	69.8	70.2	71.1	66.7	63.6	55.0	50.8	(20)
	2%	DB	54.2	56.8	64.6	73.2	80.1	85.6	87.6	86.8	78.7	69.9	58.7	54.1	(21)
		MCWB	49.5	48.7	52.4	56.9	63.9	68.9	69.4	70.1	66.4	60.8	53.0	49.7	(22)
	5%	DB	51.7	53.3	59.8	68.5	76.3	81.8	83.8	83.2	74.9	66.3	55.6	51.5	(23)
		MCWB	47.4	47.6	50.9	55.9	61.8	67.8	68.3	68.4	63.9	59.3	51.5	47.9	(24)
	10%	DB	48.4	50.0	56.0	64.6	72.3	78.3	80.0	79.3	71.2	62.7	52.9	48.5	(25)
		MCWB	45.0	45.4	49.5	53.9	60.4	66.0	66.7	67.1	62.1	57.4	49.4	45.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	51.9	56.2	61.4	69.3	72.8	72.8	73.8	69.8	64.1	56.9	52.8	(27)
		MCDB	56.0	57.8	64.7	74.0	80.1	85.0	86.1	87.3	79.3	72.1	60.3	55.5	(28)
	2%	WB	50.3	49.6	53.9	58.7	66.0	70.5	70.9	71.2	67.1	62.1	54.2	50.2	(29)
		MCDB	53.5	55.3	60.5	70.1	77.1	81.4	83.8	83.4	75.9	68.1	57.5	53.4	(30)
	5%	WB	47.5	47.7	52.1	56.7	63.6	68.8	69.3	69.6	65.1	60.2	51.9	48.1	(31)
		MCDB	51.3	52.6	58.5	66.7	73.2	79.5	81.0	80.5	72.5	65.0	55.1	51.0	(32)
	10%	WB	45.1	45.6	50.0	54.7	61.4	67.0	67.7	68.0	63.2	58.1	49.7	46.0	(33)
		MCDB	48.2	49.7	56.1	62.9	69.7	76.4	77.5	77.5	69.5	62.0	52.4	48.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	12.8	16.4	19.4	18.9	19.4	19.2	19.4	19.1	15.7	11.1	9.0	(35)
		MCDBR	12.5	18.2	24.1	27.6	27.1	25.6	26.3	26.9	25.2	20.7	15.0	11.4	(36)
	5% WB	MCWBR	9.2	11.8	14.2	14.3	12.8	11.4	10.9	11.9	13.3	12.2	10.3	8.7	(37)
		MCWBR	11.6	15.7	20.9	24.7	23.6	23.8	23.4	24.3	21.9	18.7	13.8	10.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.314	0.337	0.368	0.403	0.413	0.420	0.417	0.413	0.388	0.365	0.334	0.310	(40)
		taud	2.456	2.386	2.329	2.255	2.268	2.285	2.284	2.315	2.361	2.416	2.458	2.454	(41)
	Ebn at Noon		240	258	267	268	270	268	267	262	258	244	229	225	(42)
		Edh at Noon	20	27	34	40	41	41	40	37	32	26	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		305	547	933	1358	1601	1845	1725	1503	1136	686	352	245	(44)
	RadStd		44	93	141	191	174	154	163	151	116	61	40	43	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (91 neighbors)	N/A	N/A	N/A	N/A	+0.65	+0.54	N/A	N/A	+92	N/A

Nomenclature: See separate page